Sensorless Low-Current Start-Up Strategy of 100-kW BLDC Motor With Small Inductance

Gang Liu, Shaohua Chen, Shiqiang Zheng, and Xinda Song

***Abstract*—This paper focuses on the start-up strategy of a high-speed 100-kW brushless dc motor with small inductance using a voltage comparator. This paper includes three key technique contributions: 1) determining zero-speed positioning, 2) removing high-frequency disturbances, and 3) optimizing the controller parameters. The sensorless low-current start-up strategy is based on an amplification circuit: a low-pass filter circuit and a signal modulate circuit. The amplification circuit is used to amplify the amplitude of back electromotive force for detecting the rotor position at low speed, the low-pass filter circuit is used to remove high-frequency disturbances, and the signal modulate circuit is used to obtain the rotor position signal. A hysteresis control strategy is proposed against the load disturbance in the start-up stage. Experimental results show that the proposed sensorless start-up strategy can realize a low-current start-up.**

***Index Terms*—Amplification circuit, brushless dc motor, hysteresis control strategy, low-pass filter circuit.**

I. Introduction

A HIGH-SPEED brushless dc (BLDC) motor as a core component of the high-speed magnetic levitation (maglev) blower can be directly connected to the high-speed mechanical equipment and it effectively improves the system efficiency [1], [2]. The back-electromotive-force (EMF) technique as a sensorless strategy with profound applications has become a research focus recently, which is a scheme to estimate the rotor position indirectly by using the zero-crossing point detection from the terminal voltage [3], [4]. It has been used for wide-speed-range control of a BLDC motor [5].

There are two difficulties of the sensorless BLDC motor control strategy. The first is obtaining the position signal of the rotor when a motor is at a standstill or at a very low speed. The second is switching from an open loop to a closed loop of a BLDC motor in the start-up stage successfully. For the first condition, in [5], an offline finite-element-method-assisted position and a speed observer are proposed for a BLDC motor drive sensorless control based on the line-to-line motor flux linkage estimation. In [6], a new sensorless control method based on phase inductance vectors of a switched reluctance motor is presented. In [7], a start-up method for reducing the starting current of a permanent-magnet synchronous motor is proposed. Minimum current start-up is achieved by combining two position-sensorless control methods that require no harmonic injection. In [8] and [9], the main factors that influence the instantaneous torque performance of a small-inductance BLDC motor are analyzed. For the second condition, in [10], a characteristic of the measured terminal voltage that is used to detect the rotor position of BLDC motors controlled with a 120° conduction method has been analyzed. In [11], in the phase voltage method- or terminal voltage method-based sensorless BLDC motor drives, armature reaction may cause a rotor position detection error. In [12], winding factor calculation is discussed for the modeling and analysis of an air-core permanent magnet linear synchronous motor. A closed-form equation for back-EMF waveform and its harmonic content can be found by the proposed method for concentrated and distributed windings. In [13], a new sensorless method for estimating the speed and position of brushed dc motors using space vector modulations has been presented. The method uses sensorless techniques based on the ripple component. In [14], a sensorless control based on a hysteresis comparator of terminal voltage and a potential start-up method with a high starting torque for an automotive fuel pump application has been presented. Some approaches use the zero-crossing points of three-phase line-to-line voltages, which can directly obtain the point without the center line of the motor [15]–[17].This method has been directly compared with the electric potential of the line and the phase shifter is omitted; the structure is simple and easy to implement. However, the high-frequency noise signal is easy to include. The zero-crossing point of the back-EMF for generating proper commutation control of the inverter is calculated by sampling the voltage of the floating phase without using current and position sensors [18]–[20]. In [21], a method of pulse width modulation (PWM) raising the frequency and voltage of a driving circuit to solve the problem of zero-rotation speed has been presented. The procedure is to excite two phases of the three-phase windings for a preset time; this will pull the rotor to a fixed position, and then gradually increase the voltage and frequency of the motor.

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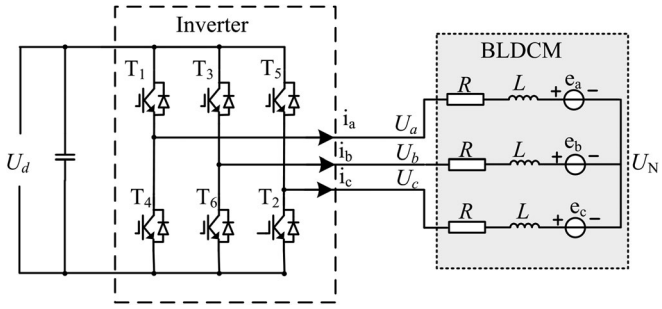


Fig. 1. Block diagram of a BLDC motor drive system.

Most sensorless BLDC motor control techniques are based on phase voltage, current, etc. However, when a motor is at a standstill or at a very low speed, it is well known that the signal is too small or contains too many noise signals to estimate a precise rotor position. Especially, the start-up of a sensorless high-energy density BLDC motor with small inductance becomes particularly difficult. Therefore, designing a low-cost and high-reliability start-up method is necessary.

In this paper, a new sensorless control strategy of a 100-kW BLDC motor is proposed. In hardware, the back-EMF signal detection circuit is proposed to remove high-frequency disturbances and obtain the rotor position signal at low speed, which is based on an amplification circuit: a low-pass filter circuit and a signal modulate circuit. In software, the hysteresis control strategy is proposed against the load disturbance in the start-up stage. Experimental results show that the proposed sensorless start-up strategy can realize high-reliability start-up.

The rest of this paper is organized as follows. The issues arising in a traditional start-up method are described in Section II. The new sensorless control strategy is presented in Section III. The implementation and experimental evaluation of this control strategy are presented in Section IV. Finally, this paper concludes in Section V.

II. Starting State Model of a BLDC Motor

A. Mathematical Model of a BLDC Motor

The block diagram of BLDCM drive system is shown in Fig. 1.

The block diagram shows the drive circuit structure of a high-speed BLDC motor, where U_N is the output voltage of the rectifier; e_a, e_b, and e_c are the back-EMFs; R is the phase-winding resistance; and L is the self-inductance.

The stator voltage equations of a BLDC motor can be given as

$$\begin{bmatrix} U_a \\ U_b \\ U_c \end{bmatrix} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} L & 0 & 0 \\ 0 & L & 0 \\ 0 & 0 & L \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \tag{1}$$

where e_a, e_b, and e_c are the EMFs; i_a, i_b, and i_c are the three-phase currents; U_a, U_b, and U_c are the three-phase voltages; R is the stator resistance; and L is the self-inductance.

Considering, for example, the two phases A and B when the ON-PWM is used, the nonconduction phase current flow is small enough to be neglected:

$$e_a = -e_b = E, i_a = -i_b \tag{2}$$

$$\Delta U = U_d = 2Ri_a + 2L\frac{di_a}{dt} + 2E. \tag{3}$$

When the high-frequency PWM is modulated, the stator resistance drop is small enough to be neglected:

$$\frac{di_a}{dt} = \frac{U_d/2 - E}{L} \tag{4}$$

$$\Delta U = 2Ri_a + 2L\frac{di_a}{dt} + 2E = 0. \tag{5}$$

Because the maglev motor is very small, it can be neglected

$$\frac{di_a}{dt} = -\frac{E}{L}. \tag{6}$$

In a PWM switching period, the average current is

$$\begin{aligned} I_{a1} &= \frac{1}{T}\left[\int_0^{DT}\left(\frac{\Delta U/2 - E}{L}t + i_{a1\min}\right)dt \right. \\ &\quad \left. + \int_{DT}^{T}\left(-\frac{E}{L}t + \frac{\Delta U}{2L}DT + i_{a1\min}\right)dt\right] \\ &= \frac{\Delta U}{2L}\left(D - \frac{1}{2}D^2\right)T - \frac{E}{2L}T + i_{a1\min} \end{aligned} \tag{7}$$

where D is the duty cycle of PWM, T is the PWM modulation time period, and $i_{a\min}$ is the minimum value of steady current.

Motor electromagnetic torque is

$$T_e = \frac{2E}{\omega_m}\left(\frac{\Delta U}{2L}\left(D - \frac{1}{2}D^2\right)T - \frac{E}{2L}T + i_{a1\min}\right). \tag{8}$$

B. Analysis of Phase Detection at Low Speed

The traditional rotor position testing technique used the back-EMF as the signal of the rotor position cannot test the exact rotor position at start-up and the current is too large. The electromagnetic torque T_e is regarded as a known input and is calculated by the following equation:

$$T_e = \frac{1}{\omega_m}(e_a i_a + e_b i_b + e_c i_c) = \frac{2E}{\omega_m} i_a \tag{9}$$

where T_e is the electromagnetic torque, ω_m is the angular velocity, and E is the amplitude of back-EMF. The ratio of the back-EMF amplitude and the angular velocity E/ω_m is a certain value. From (9), it can be observed that electromagnetic torque increases with increase of current; when the electromagnetic torque is greater than the load torque, the motor starts to rotate, and the starting process is completed. The equivalent circuit of motor start-up is shown in Fig. 2.

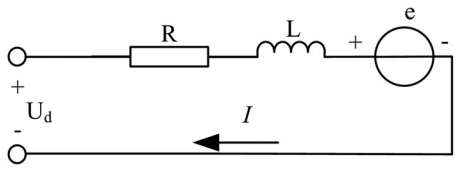


Fig. 2. Equivalent circuit of motor start-up.

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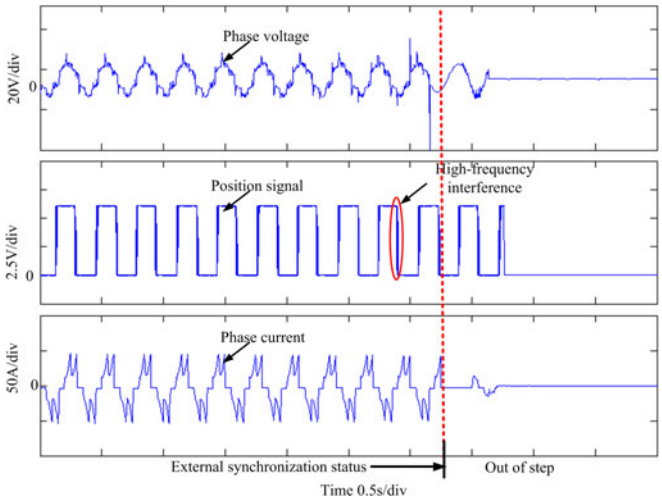


Fig. 3. Phase voltage, position signal, and phase current in the traditional "three-stage" starting stage.

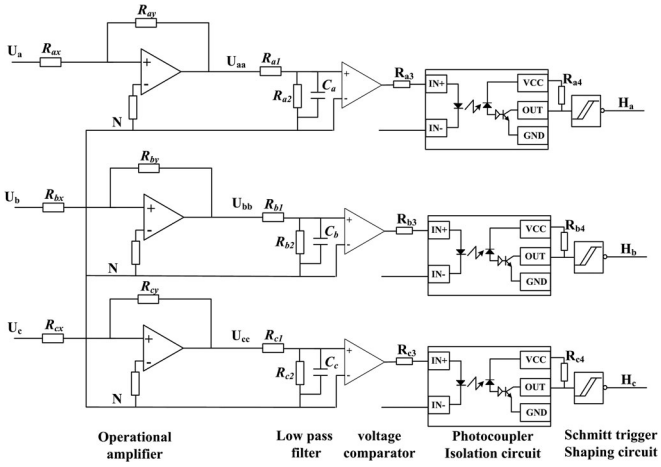


Fig. 4. Diagram of a sensorless commutation signals detection circuit.

The equivalent current equation can be given as

$$|I| = \frac{U_d - |e|}{\sqrt{R^2 + (\omega L)^2}} \tag{10}$$

where U_d is the winding-voltage of motor stator, L is the equivalent inductance, R is the equivalent resistance, ω is the angular velocity, and e is the equivalent back-EMF.

When the motor is at a standstill or at a very low speed, the back-EMF $e \approx 0$ and the angular velocity $\omega \approx 0$. Taking the experimental motor in this paper as an example, $U_d = 540\,\mathrm{V}$, $R \approx 9 \times 10^{-4}\,\Omega$, the modulation frequency is 5 kH, the minimum duty cycle is 5%, the amplitude of instantaneous current is $|I| \approx 6 \times 10^5$ A, and the large start-up current will burn out the power device. At the same time, in the traditional "three-step start-up" stage, the phase voltage waveform distortion is serious, the amplitude of phase current is too large, and the bus voltage adopts traditional PWM modulation. This method is sensitive to the motor parameters, load, and switching speed. The high-frequency interference signal will cause the motor to be out of step. Therefore, it is needed to design the start-up control strategy, as shown in Fig. 3.

III. New Sensorless Control Strategy of the BLDC Motors

When the motor is at a standstill or at a very low speed, the back-EMF is too small to estimate a precise rotor position. Therefore, a specific circuit of position detection and compensation is designed. Fig. 4 shows the block diagram of a sensorless control by using a voltage comparator to obtain rotor position signal at low speed. It consists of the operational amplifier for amplifying the amplitude of the starting back-EMF, low-pass filters (LPFs) for suppressing the high switching frequency ripples, optical couplers for isolating interference signals, and Schmitt trigger shaping circuits for increasing the output drive capability.

When the motor is at a standstill or at a very low speed, the back-EMF is fed into the operational amplifiers for amplifying the amplitude. The amplitude–frequency characteristic, phase–frequency characteristic, the cutoff frequency, and the amplitude of the voltage comparator can be given as

$$\begin{cases} |H(\omega)| = k_p \dfrac{1}{\sqrt{(R_{a1}/R_{a2}+1)^2 + (R_a \cdot \mathrm{C}_a \cdot 3\omega)^2}} \\ \psi(\omega) = -\arctan\left(\dfrac{R_{a1} \cdot \mathrm{C} \cdot 3\omega}{R_{a1}/R_{a2}+1}\right) \\ f_o = \dfrac{R_{a1}+R_{a2}}{2\pi \cdot R_{a1} \cdot R_{a2} \cdot C} \\ V_m = k_p \cdot \dfrac{R_{a2}}{R_{a1}+R_{a2}} \cdot \dfrac{K_e \omega_0}{p} \end{cases} \tag{11}$$

where $k_p = \frac{R_{ay}}{R_{ax}}$ is the amplificatory coefficient of the circuit, V_m is the amplitude of the voltage comparator, K_e is the coefficient of back-EMF, ω_0 is the angle of cutoff frequency, and p is the pole pairs of a motor.

When the motor speed is very low, the amplitude of the voltage comparator is too low to detect an effective signal. Therefore, the amplifier circuit is used to improve the detection sensitivity of back-EMF at low speed of the motor. With increase of the rotational speed, the amplitude of the counter-EMF increases gradually, when the transfer frequency $f > f_0$, the low-pass filter inertia link plays a major role, and the amplitude of the voltage comparator is limited by the inertia of the constraint. According to the experimental object, the selections of the constraints can be given as

$$\begin{cases} E_m \geq 300 \\ P_{a2} = \dfrac{k_p^2 E_m^2}{2R_{a2}} < 0.5 \\ \psi(\omega) = -\arctan\left(\dfrac{R_{a1} \cdot C \cdot 3\omega}{R_{a1}/R_{a2}+1}\right) \geq -90 \\ f_o = \dfrac{R_{a1}+R_{a2}}{2\pi \cdot R_{a1} \cdot R_{a2} \cdot C} \leq 5 \\ 2 \leq k_p \leq 10 \\ 0.04 \leq V_m \leq 0.93. \end{cases} \tag{12}$$

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TABLE I
VOLTAGE COMPARATORS

R_{a1}	R_{a2}	C_a	f_0	k_p	p	K_e	V_{m_min}
480 kΩ	910 kΩ	0.1 μf	4.98 Hz	3	2	0.01	0.049

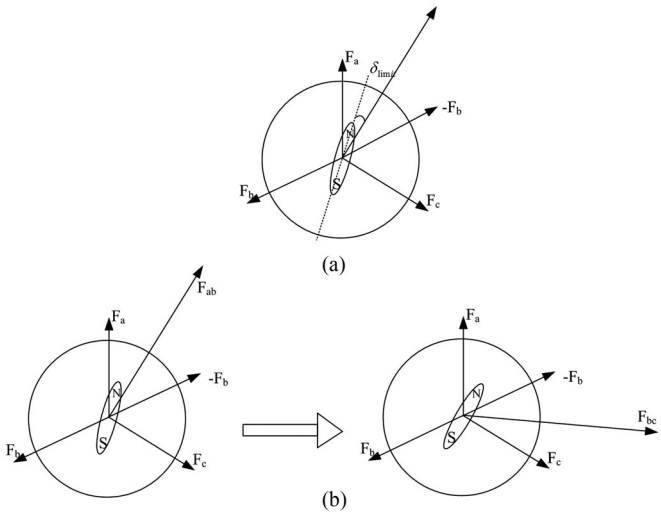


Fig. 5. Synthetic magnetic moment and rotor position by a two-step positioning method.

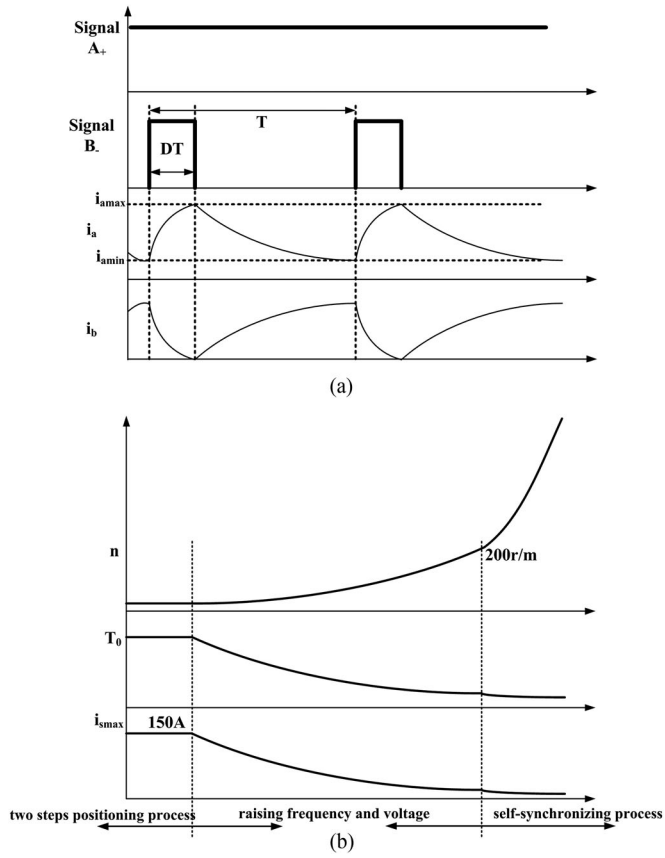


Fig. 6. Three-phase switch state, phase current, load torque, and maximum starting current in the start-up process: (a) three-phase switch state and phase current and (b) the load torque and maximum starting current.

In this paper, the maximum amplitude of back-EMF $E_m = 300$, P_{a2} is the maximum power of resistance, and the range of V_m is determined by the effective voltage of the voltage comparator. The voltage comparators are shown in Table I.

In the start-up stage, the electromagnetic torque must overcome the inherent torque so that the motor can be rotated to the position. The initial position of the rotor is random, and the rotor position can be rotated to a position with a large current when the position of the rotor is small; if there is no current limit, it is easy to damage power devices. The rotor position and magnetic moment are shown in Fig. 5(a). The angle of the limit is defined as follows:

$$\delta_{\text{limit}} = \sin^{-1} \frac{2T_0}{3pI_{s\max}K_T} \tag{13}$$

where T_0 is the inherent torque of the motor, $I_{s\max}$ is the phase current amplitude, p is the motor pole number, and K_T is the torque coefficient.

In the start-up stage, if the rotor position is in the range $-\delta_{\text{limit}} < \delta < \delta_{\text{limit}}$ or in $\pi - \delta_{\text{limit}} < \delta < \pi + \delta_{\text{limit}}$, the starting current will become very large until the rotor is pulled to a certain position, and it is easy to damage power devices. In order to solve this problem, in this paper a two-step rotor preorientation method is proposed. With the two successive electromagnetic torques that are 60° apart to ensure that the rotor is pulled to the specified position, the rotor is positioned quickly, as shown in Fig. 5(b).

The max starting current can be given as

$$I_{s\max} = \frac{2T_0}{3pK_T \sin\delta_{\text{limit}}} \tag{14}$$

where T_0 is the inherent torque of the motor, $I_{s\max}$ is the phase current amplitude, p is the motor pole number, and K_T is the torque coefficient.

In order to limit the starting current, the starting current threshold is set up. For the objective of this paper, $p = 4$, $K_T = 0.168$, $T_0 = 2$, and attention is paid to the safety of the equipment so the maximum starting current is set to 150 A; with the increase of the motor speed, the load torque and the current threshold become small. As for the experimental motor in this paper, the positioning of the blind area is less than 1° and the switching speed of the open-loop start-up to the self-synchronizing state is $100\,\text{r/min}$. Taking the two phases A and B as an example, the load torque, maximum starting current, three-phase switch state, and phase current are shown in Fig. 6.

As shown in Fig. 6(a), in one cycle, A_+ keeps the conduction state and B_- is in the PWM switch state; by limiting the current threshold, the current value is limited under the condition that the output torque is greater than the inherent torque of the motor to ensure the safety of the motor start-up.

At the start-up stage, the inherent torque is the largest due to the weight of the rotor itself. The rotor inertia force

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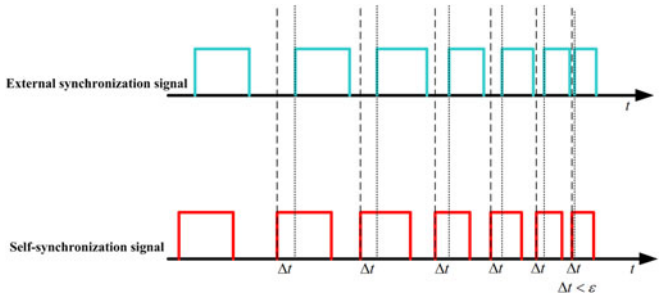


Fig. 7. External synchronization signal and self-synchronization signal.

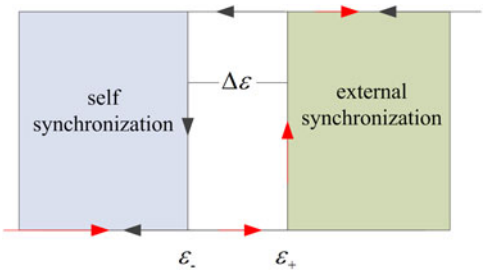


Fig. 8. Hysteresis loop schematic diagram.

increases with increase of speed, then the inherent torque begins to decrease. This paper realizes the open-loop start of the motor using DSP and extracts the external synchronization signal by the given frequency. When the back-EMF becomes large enough to be detected, compare the external synchronization signal with the self-synchronization signal. As shown in Fig. 7, when the difference between the external synchronization signal and self-synchronization signal is less than a specific value ε, the synchronization stage switches from external synchronization to self-synchronization and the sensorless "three-step startup" control of a BLDC motor is achieved.

The transition process between external synchronization and self-synchronization is achieved by a hysteresis controller. The hysteresis loop schematic diagram is shown in Fig. 8. The hysteretic controller can be expressed as

$$\chi(\Delta t) = \begin{cases} 1 & (\Delta t > \varepsilon_+) \\ 0 & (\Delta t < \varepsilon_-) \end{cases}. \tag{15}$$

As shown in Fig. 8, the region of Δt is divided into three sections. If Δt is greater than ε_+, the control mode is in an external synchronization stage, whereas if Δt is smaller than ε_-, the control mode is in the self-synchronization stage. In this paper, ε_- is 8 ms and ε_+ is 10 ms. The proper error time of ε_- and ε_+ is selected from calculation and many initial experiments, to ensure that the switching speed is less than 200 r/min.

As shown in Fig. 9, the closed-loop control system contains a speed controller, a current controller, a commutation logic control link, a signal detection link, a signal processing link, etc. On the basis of the usual BLDC motor speed control system, a closed-loop sensorless commutation algorithm is increased. The system flow chart is shown in Fig. 10.

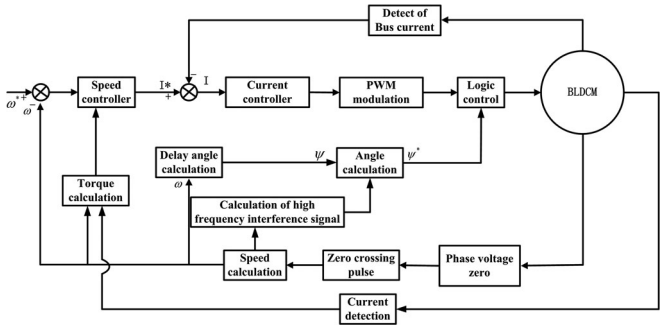


Fig. 9. Closed-loop control system.

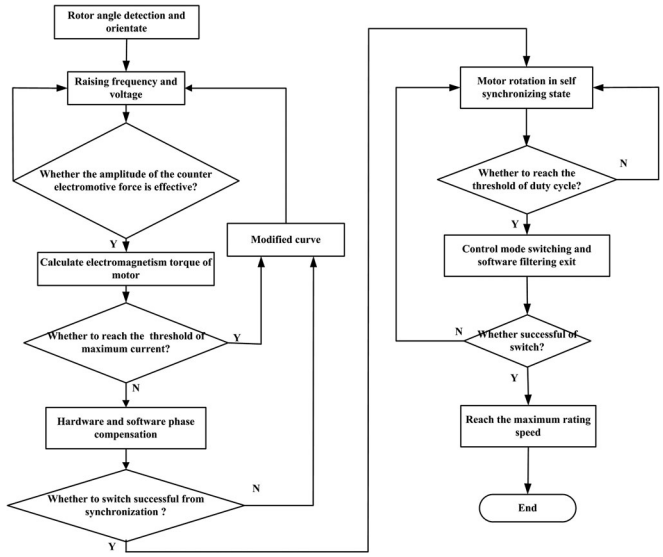


Fig. 10. System flow chart.

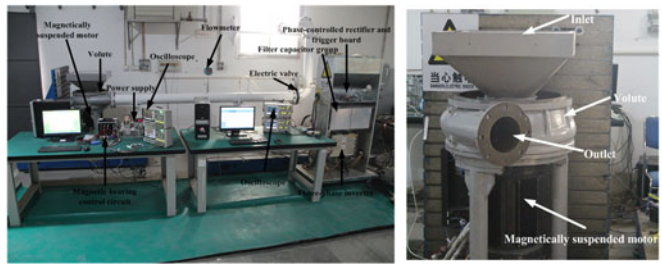


Fig. 11. Experimental platform of a high-speed maglev blower and an air blower.

IV. Experimental Verification

On the basis of the above analysis, an experimental platform can be built by using a processor having a TMS320LF28335 control chip, with a clock frequency 40 MHz and PWM modulation frequency of 10 kHz, and an inverter using an Infineon 6PS18012E4FG35689 module. An experimental platform and an experimental blower are shown in Fig. 11. Motor parameters are given in Table II.

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TABLE II
SPECIFICATION OF THE EXPERIMENTAL BLDC MOTOR

Motor parameters Surface mounted BLDC motor	
Rated dc voltage	480 V
Rated current	250 A
Rated speed, n	20 000 r/min
Number of poles, P	4
Phase resistance, R	1.3 mΩ
Phase inductance, L	0.051 mH

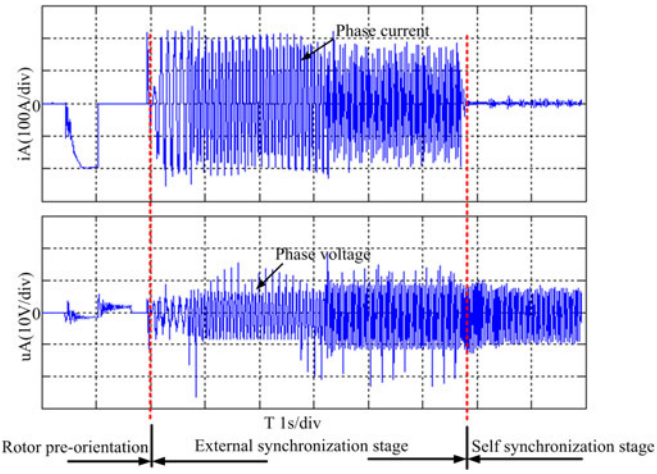


Fig. 12. Phase current and phase voltage waveforms in the traditional "three-stage" starting stage.

For the purpose of a contrast analysis, this paper analyzed the traditional "three-step startup" method. Fig. 12 shows the phase current and phase voltage waveforms without the voltage comparator. Because the back-EMF cannot be detected at a low speed, the speed of the motor switching from the external synchronization to the self-synchronization needs to be raised to 360 r/min or above. The phase current peak is 200 A, until switching to the self-synchronization stage. When the starting time is about 8 s, the starting curve is difficult to fit and the problem of starting becomes more difficult. The phase current in the external synchronization state is far greater than the phase current in the self-synchronization state without the current threshold control. The large phase current can easily damage the power device, which may cause noise and vibration. The parameters of the low-pass filter are very important for the suppression of high-frequency noise when the motor starts. Fig. 13 shows the phase current, phase voltage, and position signal waveforms in the new control strategy; at the switching time, the motor speed is 280 r/min, the phase current peak is 100 A, and the starting time is about 5 s.

The starting curves in the traditional method and the new control strategy are shown in Fig. 14. As shown in Fig. 14(a), the motor switching speed of the conventional method is 1.25 times larger than the motor switching speed of the proposed method; the starting time of conventional method is two times larger than the starting time of the proposed method. The torque curve

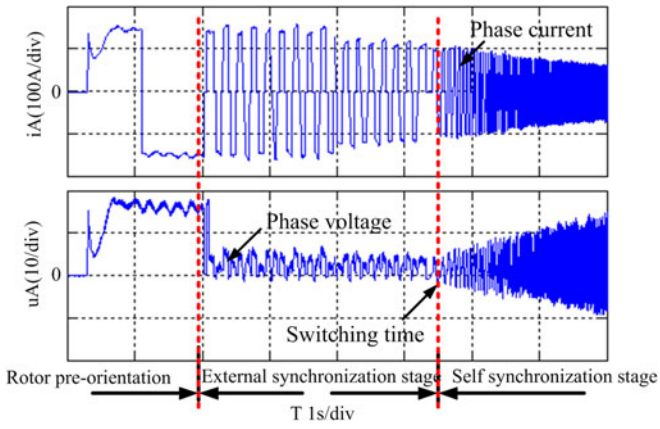
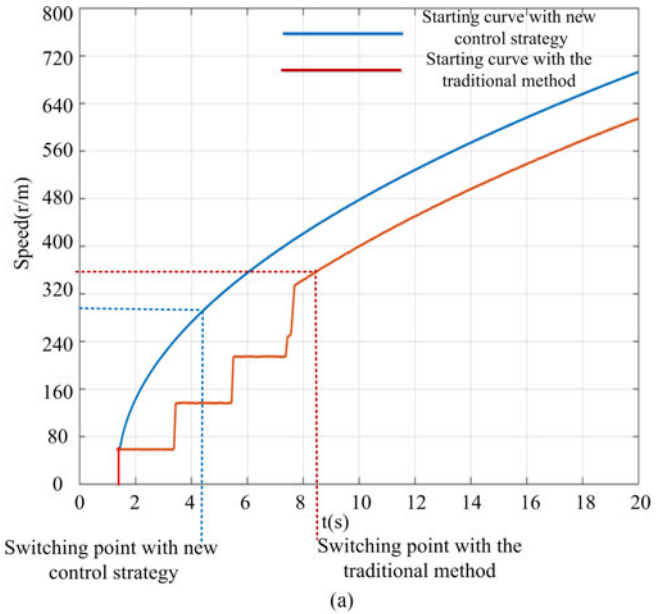


Fig. 13. Phase current, phase voltage, and position signal waveforms in the new control strategy.



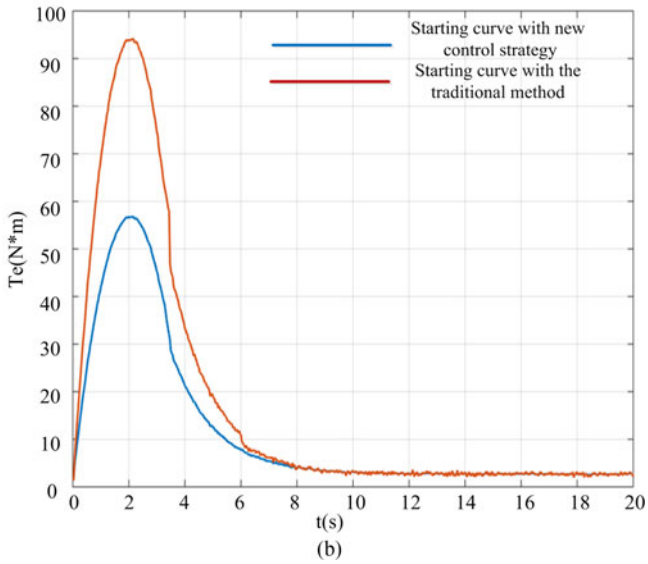


Fig. 14. Starting curve in the traditional method and the new control strategy: (a) speed curve and (b) torque curve.

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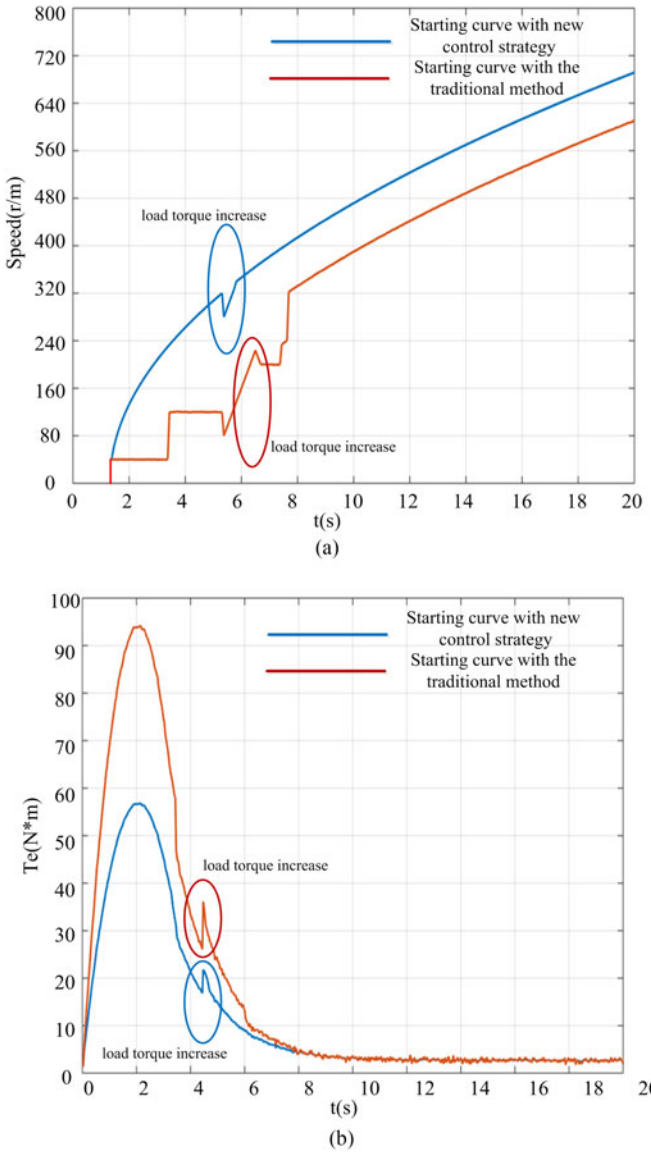


Fig. 15. Starting curve in the traditional method and the new control strategy when the motor load is increased by about 20%: (a) speed curve and (b) torque curve.

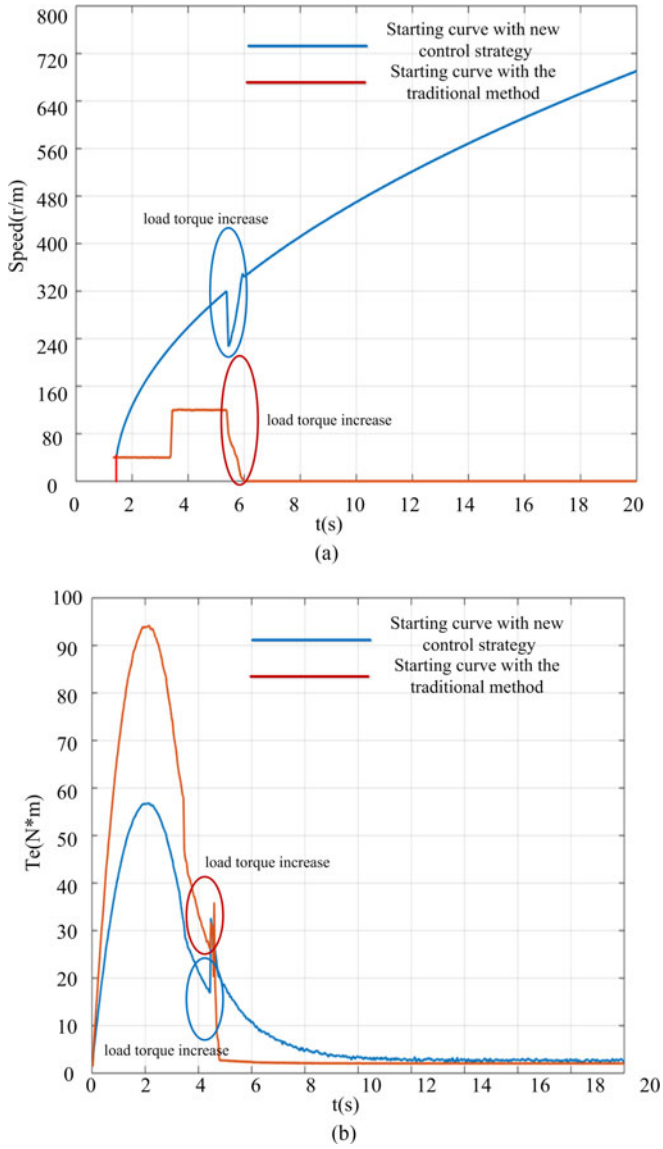


Fig. 16. Starting curve in the traditional method and the new control strategy when the motor load is increased by about 40%. (a) speed curve and (b) torque curve.

is shown in Fig. 14(b). The starting torque of the conventional method is 1.7 times larger than the starting time of the proposed method, and the starting current of the conventional method is also larger than the starting time of the proposed method.

In order to verify the effectiveness of the control strategy on the system load changes, when the motor starts, the motor load suddenly changes. Fig. 15 shows the speed curve in the start-up stage when the load torque increased by about 20%; as shown in Fig. 15(a), the speed reduces when the load torque increases both in the conventional method and in the new control strategy. As the increment is less the motor can start up; the torque curve is shown in Fig. 15(b). The starting curves in the traditional method and the new control strategy when the load torque increased by about 40% in the starting stage are shown in Fig. 16. The speed can respond rapidly to a new load torque in the new control strategy and the closed loop can quickly respond to increase in output torque in the new control strategy. However, for the traditional method, the disturbance torque is beyond the traditional method control range and it causes the motor to be out of step. The torque curve is shown in Fig. 16(b).

Fig. 17(a) shows the phase voltage and phase current waveforms when the motor load is suddenly changed in the traditional "three-step start-up" stage. Fig. 17(a) shows the phase voltage and phase current waveforms when the motor load is suddenly increased. The speed of the motor is $100\,\mathrm{r/min}$ and the traditional "three-step start-up" stage cannot adjust the switch state in time, which leads to the motor to be out of step and its failure to start. Fig. 17(b) shows the process details of the phase voltage and the phase current waveform when the motor load is suddenly increased; it causes the motor to be out of step. The

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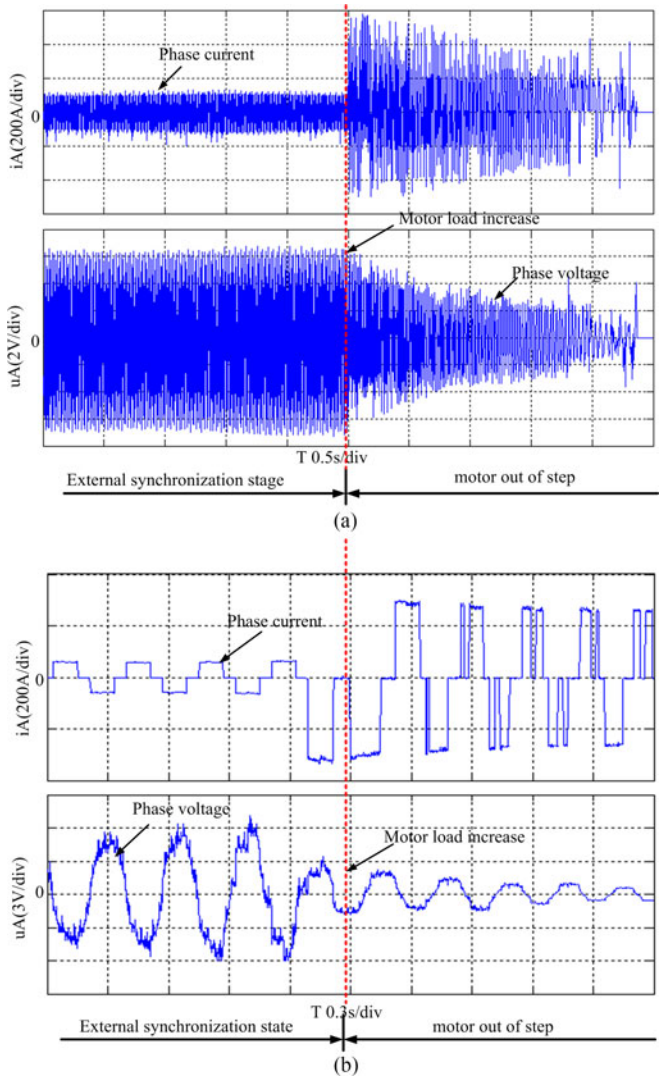


Fig. 17. Phase voltage and phase current waveforms when the motor load is suddenly increased in the traditional "three-stage" start stage. (a) Phase voltage and phase current waveforms when the motor load is suddenly increased. (b) Process details of the phase voltage and phase current waveforms when the motor load is suddenly increased.

traditional "three-step start-up" has no coping capacity for the suddenly increased motor load.

Fig. 18 shows the motor phase voltage and phase current waveforms when the motor load is suddenly reduced. The starting time is about 6 s.

As shown in Fig. 19, when the motor load is suddenly increased, the motor phase current increases and the motor speed decreases; if it is lower than the threshold speed, the phase voltage is not sufficient to achieve the position signal, the motor is still in the external synchronization stage, until the threshold speed is reached, and the self-synchronization switch is realized. When the motor load is suddenly decreased, the motor phase current decreases and the motor speed increases. When the threshold speed is reached, the position signal is extracted by the phase voltage and the self-synchronization switch is realized, as shown in Fig. 20. Experiments show that the control strategy is robust in the start-up phase.

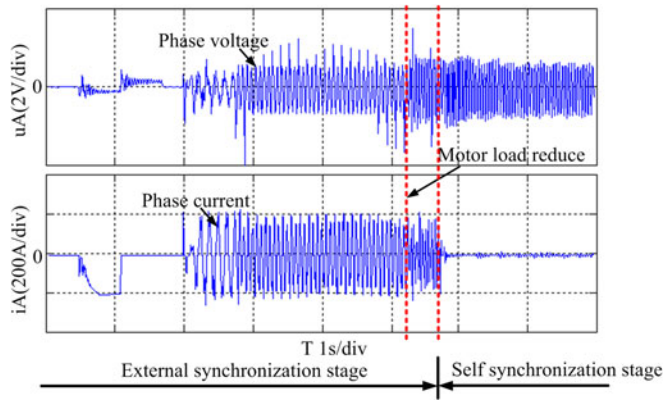


Fig. 18. Phase voltage and phase current waveforms when the motor load is suddenly reduced in the traditional "three-stage" start stage.

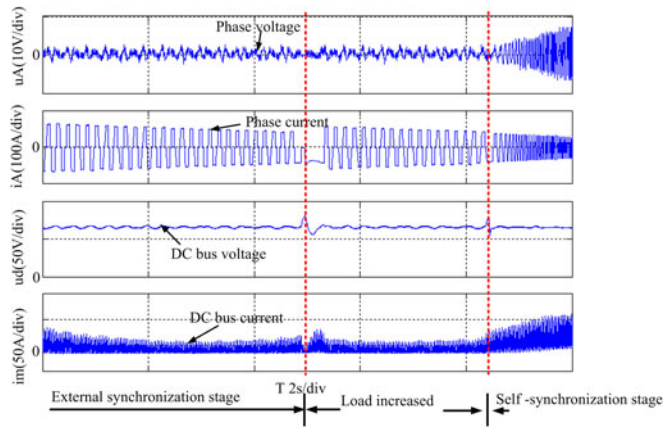


Fig. 19. Phase voltage and phase current waveforms when the motor load is suddenly increased in the new control strategy.

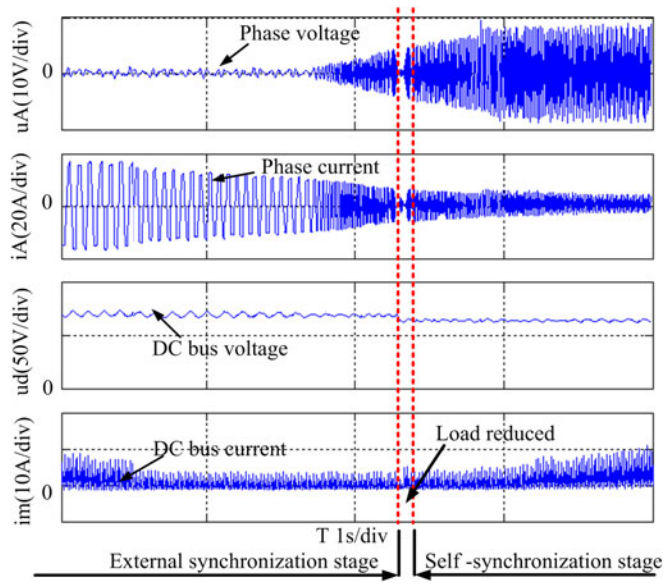


Fig. 20. Phase voltage and phase current waveforms when the motor load is suddenly reduced in the new control strategy.

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TABLE III
INDICATORS OF THE STARTING TIME IN THE TRADITIONAL METHOD AND THE NEW CONTROL STRATEGY

	Traditional method	New control strategy
Starting current	200 A	100 A
Starting time	8.5 s	4.5 s
Switching speed	360 r/min	280 r/min
Starting torque	95 N · m	55 N · m

The main indicators of the starting time are compared in Table III.

V. CONCLUSION

This paper develops the start-up strategy of a high-speed 100-kW BLDC motor with small inductance. Two key techniques of a high-speed maglev blower have been resolved:

1) Solve the problem of the high-frequency disturbances in the starting stage. The low-pass filter circuit is used to remove the high-frequency disturbances and extract the position signal accurately.
2) Solve the problem of the load disturbances in the starting stage. The closed-loop control strategy can be achieved at a very low speed, the start-up curve is very simple, and the hardware and software are of low cost and are easy to adjust.

Through the experimental results, it can be seen that the proposed sensorless and start-up techniques are ideally suited for the high-speed blower application.

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